Practice workbook · Rates of change with one chain-rule link

Linking a rate with respect to time to the rate of a length using one chain, such as dV/dt = dV/dh times dh/dt.

18 question parts: 1 worked-example retry and 17 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-CAL-14

1. Question 2 · part whole · 6 marks

2019 JUN · Paper 02

Worked-example retry · Attempt part whole only.

2 Oil is leaking from a pipe and forms a circular pool on a horizontal surface. The area of the surface of the pool is increasing at a constant rate of $8\,cm^2/s$. Find, in cm/s to 3 significant figures, the rate at which the radius of the pool is increasing when the area of the pool is $50\,cm^2$

(6)

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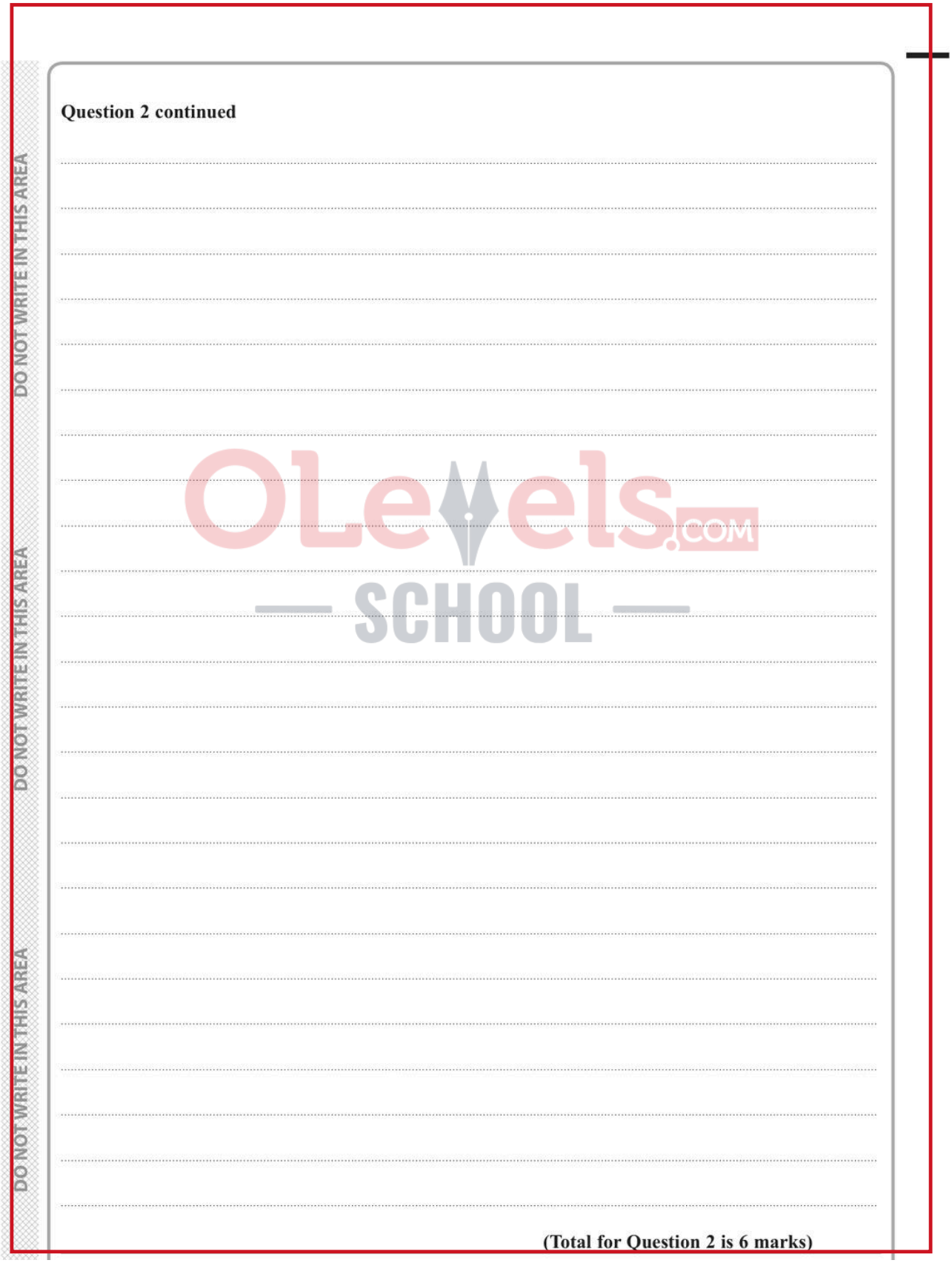
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1. Question 2 · part whole · 6 marks

2019 JUN · Paper 02

Worked-example retry · Attempt part whole only.



Question 2 continued

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(Total for Question 2 is 6 marks)

1. Question 2 · part whole · 6 marks · Working and answer

Working and answer

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2. Question 10 · part b · 3 marks

2020 NOV · Paper 01

Further practice · Attempt part b only.

10

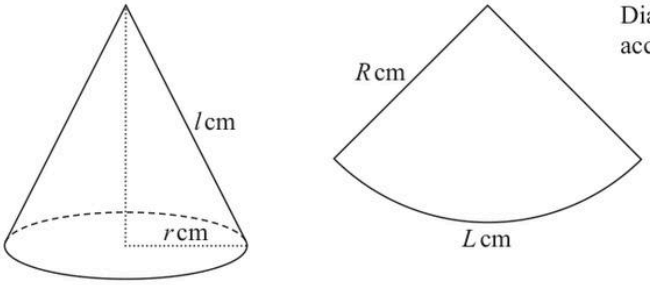


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows a right circular cone with base radius r cm and slant height l cm.
Figure 4 also shows a sector of a circle with radius R cm and arc length L cm.

The area of the curved surface of the cone is A cm^2

By considering how the sector of the circle can be folded to exactly form the curved surface of the cone with R and L suitably chosen,

(a) prove that $A = \pi rl$

(4)

Sand is poured onto a horizontal surface at a constant rate of 1.5 cm^3/s.
The sand forms a pile in the shape of a right circular cone with its base on the surface.
The curved surface area of the cone, A cm^2, increases in such a way that the height of the cone is always three times the radius of the base of the cone.

Given that $\frac{dA}{dr} = k\pi r$, where k is a constant,

(b) find the exact value of k.

(3)

(c) Calculate the rate, in cm^2/s, to 3 significant figures, at which the curved surface area of the pile is increasing when the height of the pile is 24 cm.

(5)

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26

P 6 2 2 8 0 A 0 2 6 3 2

2. Question 10 · part b · 3 marks · Working and answer

Working and answer

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3. Question 6 · part whole · 7 marks

2023 NOV · Paper 02

Further practice · Attempt part whole only.

6 The volume of oil in a container is V cm^3 when the height of the oil is h cm.
Oil is pouring into the container at a constant rate of 12 cm^3/s.
Given that $V = 3h^3$

find the exact rate, in cm/s, at which the height of the oil is increasing when $V = 1536$ cm^3

(7)

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P 7 3 5 8 6 A 0 1 6 3 2

3. Question 6 · part whole · 7 marks

2023 NOV · Paper 02

Further practice · Attempt part whole only.

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Question 6 continued

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(Total for Question 6 is 7 marks)

3. Question 6 · part whole · 7 marks · Working and answer

Working and answer

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4. Question 7 · part whole · 6 marks

2025 JUN · Paper 01

Further practice · Attempt part whole only.

7

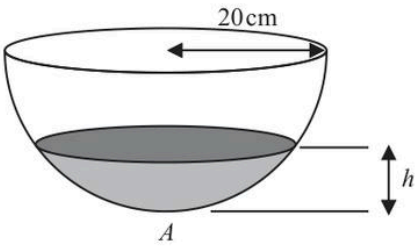


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows a hollow hemisphere with radius 20 cm

The hemisphere contains liquid, which is dripping out of a small hole at the lowest point A at a constant rate of k cm^3/s

At time t seconds after the liquid starts to drip from the hemisphere, the height of the liquid is h cm above A

The volume V cm^3 of liquid in the hemisphere is given by

$$V = \frac{\pi}{3}h^2(60 - h)$$

When $h = 12$, the height of the liquid is decreasing at a rate of $\frac{1}{60}$ cm/s

Find the value of k
Give your answer in terms of π

(6)

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4. Question 7 · part whole · 6 marks

2025 JUN · Paper 01

Further practice · Attempt part whole only.

Question 7 continued

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4. Question 7 · part whole · 6 marks

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Further practice · Attempt part whole only.

Question 7 continued

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4. Question 7 · part whole · 6 marks

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Further practice · Attempt part whole only.

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Question 7 continued

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(Total for Question 7 is 6 marks)

4. Question 7 · part whole · 6 marks · Working and answer

Working and answer

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5. Question 3 · part whole · 6 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part whole only.

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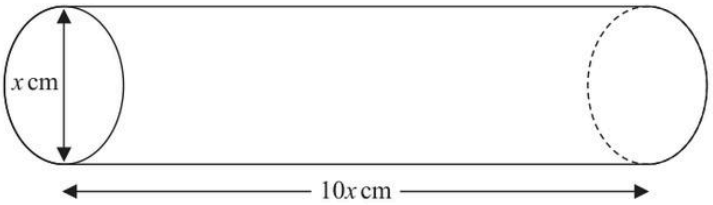


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows a solid right circular cylindrical metal rod.

The diameter of the rod is x cm and the length of the rod is $10x$ cm.

The rod is being heated so that the length of the rod is increasing at a rate of 0.005 cm/s.

Find the rate of increase, in cm^3/s to 2 significant figures, of the volume of the rod when $x = 3$

(6)

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5. Question 3 · part whole · 6 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part whole only.

Question 3 continued

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(Total for Question 3 is 6 marks)

5. Question 3 · part whole · 6 marks · Working and answer

Working and answer

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6. Question 8 · part a · 3 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part a only.

8 Liquid drips onto a large horizontal flat cloth, forming a circular stain.

The liquid starts to drip onto the cloth at time $t = 0$

The area of the stain increases at a constant rate of $1.5\,\text{cm}^2/\text{s}$

(a) Find, in terms of π, the radius of the stain at time $t = 4$ seconds.

(3)

(b) Find, in cm/s to 3 significant figures, the rate at which the radius of the stain is increasing at time $t = 4$ seconds.

(4)

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6. Question 8 · part a · 3 marks · Working and answer

Working and answer

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7. Question 8 · part b · 4 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

8 Liquid drips onto a large horizontal flat cloth, forming a circular stain.

The liquid starts to drip onto the cloth at time $t = 0$

The area of the stain increases at a constant rate of $1.5\,\text{cm}^2/\text{s}$

(a) Find, in terms of π, the radius of the stain at time $t = 4$ seconds.

(3)

(b) Find, in cm/s to 3 significant figures, the rate at which the radius of the stain is increasing at time $t = 4$ seconds.

(4)

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7. Question 8 · part b · 4 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

Question 8 continued

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(Total for Question 8 is 7 marks)

7. Question 8 · part b · 4 marks · Working and answer

Working and answer

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8. Question 6 · part whole · 6 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part whole only.

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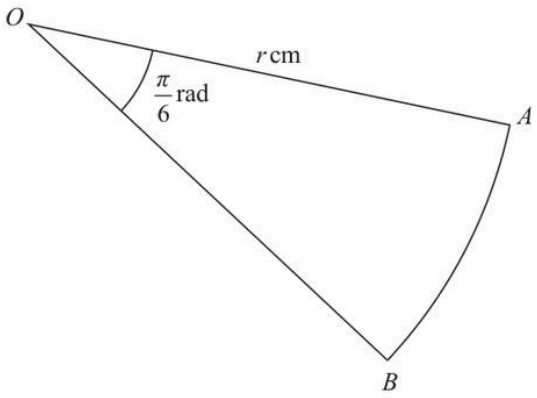


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows the sector OAB of a circle with centre O and radius r cm.

$$\angle AOB = \frac{\pi}{6} \text{ radians} \qquad OA = OB = r \text{ cm}$$

The area of the sector is increasing in such a way that the size of $\angle AOB$ remains constant, and the lengths OA and OB are both increasing at a constant rate of 0.2 cm/s

Find the exact rate of change, in cm^2/s, of the area of the sector when the length of arc AB is $\frac{5\pi}{2}$

(6)

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8. Question 6 · part whole · 6 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part whole only.

Question 6 continued

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(Total for Question 6 is 6 marks)

8. Question 6 · part whole · 6 marks · Working and answer

Working and answer

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9. Question 4 · part whole · 8 marks

2024 JUN · Paper 01

Further practice · Attempt part whole only.

4 The surface area of a sphere with radius r cm is increasing at a constant rate of 50π cm^2/s

Find, in cm^3, the exact volume of the sphere at the instant when the rate of increase of r is $\frac{5}{12}$ cm/s

(8)

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9. Question 4 · part whole · 8 marks

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Further practice · Attempt part whole only.

Question 4 continued

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(Total for Question 4 is 8 marks)

9. Question 4 · part whole · 8 marks · Working and answer

Working and answer

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10. Question 5 · part whole · 6 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part whole only.

5 The force F newtons between two magnetic poles is given by the formula

$$F = \frac{3}{20r^2}$$

where r is the distance, in centimetres, between the poles.

The distance between the two poles is increasing at a constant rate of 0.7 cm/s

Find the rate of change of F, in newtons/s to 3 significant figures, when the distance between the poles is 2.8 cm

(6)

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10. Question 5 · part whole · 6 marks

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Further practice · Attempt part whole only.

Question 5 continued

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(Total for Question 5 is 6 marks)

10. Question 5 · part whole · 6 marks · Working and answer

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11. Question 9 · part a · 5 marks

2021 JUN · Paper 02

Further practice · Attempt part a only.

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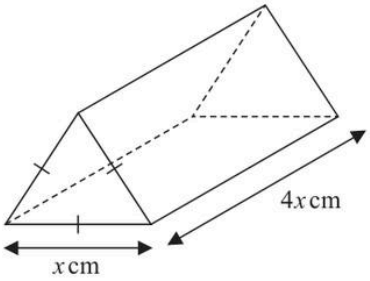


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows a metal solid S.

The solid is a right triangular prism.

The cross section of S is an equilateral triangle with sides of length x cm.
The length of S is $4x$ cm.

The prism is being heated so that the cross sectional area is increasing at a constant rate of $0.03\,\text{cm}^2/\text{s}$.

(a) Find, giving your answer to 3 significant figures, $\frac{dx}{dt}$ when $x = 2$ **(5)**

(b) Find the rate of increase, in cm^3/s, of the volume of S when $x = 2$ **(3)**

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11. Question 9 · part a · 5 marks · Working and answer

Working and answer

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12. Question 6 · part b · 4 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

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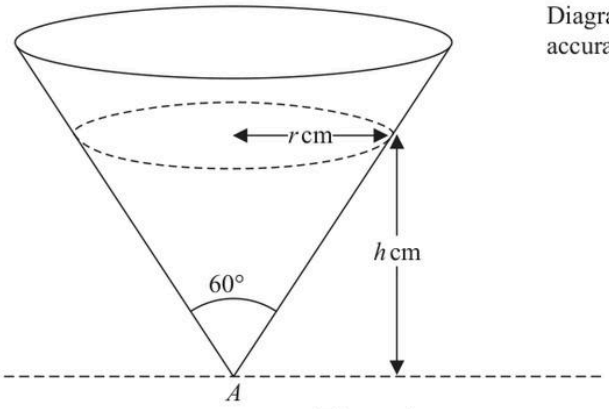


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows a hollow right circular cone fixed with its axis of symmetry vertical.

The cone is inverted and contains liquid, which is dripping out of a small hole at the vertex A of the cone at a constant rate of $0.9\,\text{cm}^3/\text{s}$.

At time t seconds after the liquid starts to drip from the cone, the height of the liquid is h cm above A. The volume of liquid in the cone at time t seconds is $V\,\text{cm}^3$

The vertical angle of the cone is $60°$

(a) Show that $V = \frac{1}{9}\pi h^3$

(2)

(b) Find, in cm/s to 3 significant figures, the rate at which the height of the liquid is decreasing when the height of the liquid in the cone above the vertex is 1.2 cm.

(4)

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12. Question 6 · part b · 4 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

Question 6 continued

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12. Question 6 · part b · 4 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

Question 6 continued

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12. Question 6 · part b · 4 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.



Question 6 continued

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(Total for Question 6 is 6 marks)

12. Question 6 · part b · 4 marks · Working and answer

Working and answer

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13. Question 6 · part whole · 6 marks

2021 NOV · Paper 01

Further practice · Attempt part whole only.

6 The volume of a sphere with radius r cm is increasing at a constant rate of $3\,\text{cm}^3/\text{s}$.

Find the rate, in cm^2/s, at which the surface area of the sphere is increasing when $r = 10$

(6)

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13. Question 6 · part whole · 6 marks

2021 NOV · Paper 01

Further practice · Attempt part whole only.

Question 6 continued

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(Total for Question 6 is 6 marks)

13. Question 6 · part whole · 6 marks · Working and answer

Working and answer

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14. Question 2 · part whole · 6 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part whole only.

2 When poured from a pipe, concrete is formed into the shape of a cuboid with a square base of side x and with a height of $3x$

The volume of the cuboid increases at a constant rate of $8\,m^3/s$

Find the rate of increase, in m/s, of x when $x = 2$ metres.

(6)

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14. Question 2 · part whole · 6 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part whole only.

Question 2 continued

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(Total for Question 2 is 6 marks)

14. Question 2 · part whole · 6 marks · Working and answer

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15. Question 6 · part b · 5 marks

2023 JUN · Paper 02

Further practice · Attempt part b only.

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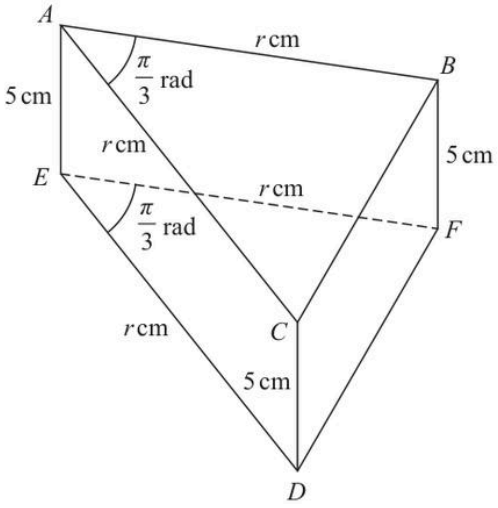


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows a right triangular prism $ABCDEF$. A cross section ABC of the prism is a triangle in which $AB = AC = r$ cm and $\angle CAB = \frac{\pi}{3}$ radians.

In the prism

$$AE = BF = CD = 5\,\text{cm} \qquad ED = EF = r\,\text{cm} \text{ and } \angle DEF = \frac{\pi}{3} \text{ radians}$$

(a) Show that the volume of the prism is $\frac{5\sqrt{3}}{4} r^2\ \text{cm}^3$

(1)

The volume of the prism is increasing in such a way that the size of $\angle CAB$ and the size of $\angle DEF$ remain constant and the length of AE, the length of BF and the length of CD remain constant.
The lengths of AB, AC, ED and EF are each increasing at a constant rate of $0.2\,\text{cm/s}$

(b) Find the exact rate of increase, in cm^3/s, of the volume of the prism when the area of the rectangular face $BCDF$ is $60\,\text{cm}^2$

(5)

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15. Question 6 · part b · 5 marks

2023 JUN · Paper 02

Further practice · Attempt part b only.

Question 6 continued

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15. Question 6 · part b · 5 marks

2023 JUN · Paper 02

Further practice · Attempt part b only.

Question 6 continued

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15. Question 6 · part b · 5 marks

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Further practice · Attempt part b only.

Question 6 continued

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(Total for Question 6 is 6 marks)

15. Question 6 · part b · 5 marks · Working and answer

Working and answer

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16. Question 5 · part whole · 5 marks

2024 NOV · Paper 02

Further practice · Attempt part whole only.

5 The height of liquid in a vessel P is h
The volume, V, of the liquid in P is given by $V = 6h^3$
Liquid is leaking from P at a constant rate of $36\,\text{cm}^3/\text{s}$

Find the exact rate of change, in cm/s, of h when $V = 384\ \text{cm}^3$

(5)

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16. Question 5 · part whole · 5 marks

2024 NOV · Paper 02

Further practice · Attempt part whole only.

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Question 5 continued

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(Total for Question 5 is 5 marks)

16. Question 5 · part whole · 5 marks · Working and answer

Working and answer

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17. Question 7 · part a · 4 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part a only.

7 The length of each side of a cube S_1 is increasing at a constant rate of 0.1 m/s.

(a) Find, in m^3/s, the rate of increase of the volume of the cube S_1 when the length of each side of the cube is 2 m.

(4)

The total surface area of a different cube S_2 is increasing at a constant rate of 0.05 m^2/s.

(b) Find in m^3/s, the rate of increase of the volume of the cube S_2 when the length of each side of the cube is 6 m.

(5)

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17. Question 7 · part a · 4 marks · Working and answer

Working and answer

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18. Question 11 · part whole · 8 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part whole only.

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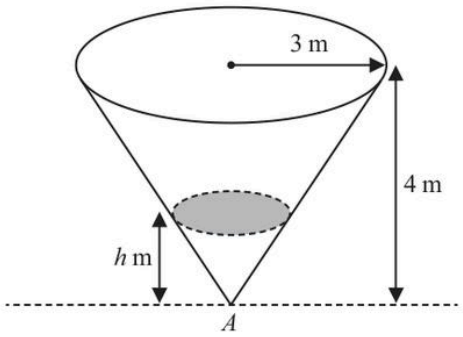


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows a hollow right circular cone with radius 3 metres and height 4 metres above the vertex A

The cone is fixed with its axis of symmetry vertical.
The cone is initially empty.

Water pours into the cone at a constant rate of 9π m^3/s
At time t seconds after the water starts to pour into the cone, the height of the water is h metres above A

Find, in m/s, the rate at which the height of the water is increasing at the instant when $h = 2$

(8)

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18. Question 11 · part whole · 8 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part whole only.

Question 11 continued

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18. Question 11 · part whole · 8 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part whole only.

Question 11 continued

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(Total for Question 11 is 8 marks)

18. Question 11 · part whole · 8 marks · Working and answer

Working and answer

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